informed decisions about continued use or replacement.

Real-time example: A manufacturer announces that a specific model of smart sensors will no longer receive security updates after a certain date. Users are advised to upgrade to newer models with ongoing security support to maintain protection against evolving threats.

By adhering to these principles, manufacturers and users can work together to create a more secure IoT ecosystem, mitigating the risks associated with connected devices and protecting sensitive data in today's ever-evolving threat landscape.

Securing home architecture with IoT

A secure smart home is a living space that leverages the convenience and security benefits of interconnected devices while prioritising data protection and minimising vulnerabilities. In a smart home, security is a top priority. Here's what it entails:

Network router and firewall: This acts as the central control point for your network traffic. A good firewall filters incoming and outgoing data, blocking unauthorised access attempts.

Smart hub (optional): This centralises communication between various IoT devices and serves as a control centre for your smart home ecosystem. Choose one with robust security features.

IoT devices: Smart lights, thermostats, locks, cameras, and more – these are the gadgets that bring intelligence to your home. Ensure each one has strong security capabilities.

Intrusion detection/prevention system (optional): These systems actively monitor your network for suspicious activity and can take steps to prevent cyberattacks.

Mobile app: This application allows you to control your smart home devices remotely. It should be secured with strong passwords and multi-factor authentication.

Amazon Timestream for IoT use cases

For IoT use cases, where there are millions of events to be processed in real-time and data feeds come from multiple IoT enabled devices and other sources, it is challenging for an architect to choose a database that can handle the throughput at par. RDBMS or NoSQL databases are difficult to manage in such cases, as they may not perform well or be too costly.

To address this, AWS has developed Timestream, which is a fast, scalable, serverless time series database that performs 1000 times faster than an RDBMS and costs one-tenth of the traditional database platforms. Timestream can store and analyse millions and trillions of events in real-time from IoT sources and can be integrated with SageMaker or QuickSight for analytics, visualisation or processing time series data.

Timestream is serverless and scalable, with an auto-scaling facility to ensure high availability with low cost and optimal resource usage. It can help manage the entire data life cycle. Timestream has a memory store for faster processing of the latest data and magnetic store for low-called historical data. Timestream provides encryption at rest and at motion, and allows key management service (KMS) based customer managed key (CMK) to encrypt historical data (in magnetic stores).

A built-in time series function enables quick analysis of time series data using SQL queries or to interpolate, aggregate and derive metrics in real-time. The data in memory operation is backed up automatically in s3 bucket for safety.

A few principles must be adopted to ensure the security of your smart home.

- **Network segmentation**: Don't let your smart fridge share the same network as your computer. Segment your network. Create a separate network (often called a 'guest network') specifically for your IoT devices. This isolates them from your personal devices, minimising damage if an IoT device is compromised.
- **Strong authentication**: Move beyond basic passwords! Implement strong and unique credentials for all your IoT devices. Consider password managers for secure storage. Additionally, enable multi-factor authentication (MFA) whenever available for an extra layer of security.
- **Secure boot and execution environment**: This ensures only authorised software runs on your devices. Imagine a bouncer at a club, verifying software before it enters the device.
- **Patch management**: Just like patching a hole in your clothes, software patches fix vulnerabilities in your devices. Set your devices to automatically download and install updates to stay protected.
- **Physical security:** Don't leave your smart gadgets vulnerable! Implement physical security measures like tamper-evident seals (e.g., smart meters) and secure enclosures for hubs or gateways.
- **User awareness**: Educate yourself and your family about potential threats and best practices. This includes recognising phishing attempts, avoiding suspicious links, and reporting anomalies to device manufacturers.

Here are two examples of how this is put into practice.

Scenario A: You purchase a smart lock.

Secure architecture: The lock connects to your secure home network with a unique password and MFA enabled for access.

Explanation: This prevents unauthorised access attempts, even if someone discovers the password.